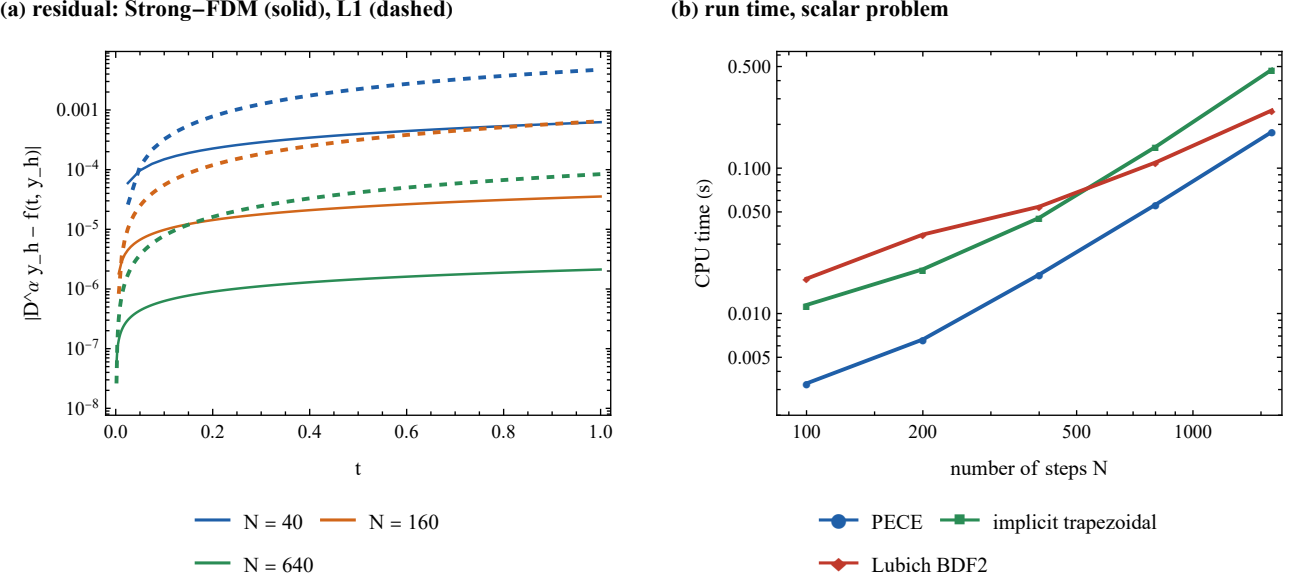

(a) residual: Strong–FDM (solid), L1 (dashed)
|D^α y_h − f(t, y_h)|
0.001
10^-4
10^-5
10^-6
10^-7
10^-8
0.0
0.2
0.4
0.6
0.8
1.0
t
N = 40
N = 160
N = 640
(b) run time, scalar problem
CPU time (s)
0.500
0.100
0.050
0.010
0.005
100
200
500
1000
number of steps N
PECE
implicit trapezoidal
Lubich BDF2